MIL-A-85097/7C(AS)
26 June 1985
SUPERSEDING
MIL-A-85097/7B(AS)
24 August 1984

MILITARY SPECIFICATION SHEET

# AIRCREW ESCAPE PROPULSION SYSTEMS (AEPS); ROCKET MOTOR MARK 84 MODS 1 AND 2

This specification is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the Mark 84 Mods 1 and 2 Rocket Motors described herein shall consist of this document and the latest issue of MIL-A-85097.

## 1. SCOPE

**1.1 Scope.** This specification sheet covers requirements unique to the Rocket Motor Mark 84 Mods 1 and 2 for use in the S-3 Aircraft.

## 2. APPLICABLE DOCUMENTS

**2.1 Government documents.** The following drawings are in addition to those listed in the cover specification, MIL-A-85097, and form a part of this specification sheet to the extent specified herein.

DRAWINGS

Naval Air Systems Command (Code Ident 30003)

| | |
|---|---|
| DL503AS102 | Rocket Motor Mark 84 Mod 1 (Seat Stabilizer) |
| DL503AS200 | Rocket Motor Mark 84 Mod 2 (Seat Stabilizer) |

## 3. REQUIREMENTS

**3.1 General.** The requirements in this specification sheet, unless in direct conflict with MIL-A-85097, shall be considered additional, not substitute. The Mark 84 Mod 1 Rocket Motor shall conform to DL503AS102 and the Mark 84 Mod 2 Rocket Motor shall conform to DL503AS200.

**3.2 Preproduction.**

**3.2.1 Propellant aging.** When tested in accordance with MIL-A-85097, the propellant shall meet the limit requirements specified by the contractor and which are acceptable to the Government.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Engineering Center, Systems Engineering & Stdzn. Department (SESD) Code 93, Lakehurst, NJ 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.



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FSC 1377

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**3.3 Preproduction/production.**

**3.3.1 Propellant.** The propellant shall conform to the contractor's in-house propellant specification when tested for data specified in MIL-A-85097. The contractor shall make available the results of the tested values and the in-house specification limits for comparison (see 6.2.2).

**3.3.2 Ballistic requirements.** The ballistic requirements shall be as specified in table I and provide the additional data as specified herein.

**3.3.3 Unstable burning.** Unstable burning does not apply to the Mark 84 Mods 1 and 2 Rocket Motors.

**3.3.4 Sear.** The force required to withdraw the sear shall be not less than 25 pounds nor greater than 35 pounds. Travel of sear to actuate the rocket motor shall be not less than 0.25 inch nor greater than 0.69 inch. Travel to completely remove the sear shall be 0.75 inch maximum.

TABLE I. Ballistic performance requirements.

| Parameter | −40° ± 5°F | | 160° ± 5°F | |
|---|---|---|---|---|
| | Min | Max | Min | Max |
| Max. time from zero time to 70% peak thrust (sec) | - | .030 | - | .030 |
| Peak thrust (lb) | 550 | 980 | 550 | 980 |
| Ignition peak thrust (lb) | - | - | - | 1350 |
| Burn time (sec) | .331 | .424 | .230 | .305 |
| Action time (sec) | .352 | .480 | .278 | .323 |
| Impulse (lb-sec) | 220 | 244 | 220 | 249 |
| Torque about longitudinal axis (in-lb) | −6 | +6 | −6 | +6 |

**3.3.4.1 Sear pull test.** Sear pull tests shall be conducted on all rocket motors. The sear force shall be statically determined, before final assembly, by means of a pull test fixture. The sear pull velocity shall be 18 ± 2 ft/sec. Test results shall be made available to the procuring activity with DCASR acceptance.

**3.3.5 Center of gravity.** The calculated or measured center of gravity shall be within 0.03 inch of the axis of rotation.

## 4. QUALITY ASSURANCE PROVISIONS

**4.1 Preproduction/production.**

**4.1.1 Conditioning temperatures.** The conditioning temperatures and tolerance shall be as follows:

a. Cold: −40° ± 5°F
b. Hot: 160° ± 5°F

**4.1.2 Conditioning periods.** Time for conditioning prior to static firing or during temperature cycling shall be as follows:

a. Static firing: 4 hours, minimum
b. Temperature cycling: 4 hours, minimum

**4.1.3 Radiographic inspection.** All components or assemblies shall be radiographed in two views 90° apart.



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## 5. PACKAGING

(This section is not applicable to this specification sheet).

## 6. NOTES

6.1 **Intended use.** The Mark 84 Mod 1 and Mod 2 Rocket Motors are used as a means of stabilizing the ejection seat during an emergency escape of personnel from the S-3A Aircraft.

6.2 **Ordering data.**

6.2.1 **Acquisition requirements.** Acquisition documents should specify the following:

a. Title, number and date of this specification
b. Mark and Mod designation of rocket motor

6.2.2 **Data requirements.** When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL), the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of FAR 52-209-3 and -4 are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs.

| Paragraph no. | Data requirement title | Applicable DID no. | Option |
|---|---|---|---|
| 3.3.1 | Acceptance Test Report | UDI-T-22741A | - |
| 3.3.4.1 | Test Report | DI-T-3718A | - |

(Data item descriptions related to this specification, and identified in this section will be approved and listed as such in DoD 5000.19L., Vol.II, AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.3 **Definitions.**

6.3.1 **Time from zero time to 70% peak thrust.** Time to 70% peak thrust is the interval between the ignition signal (sear pin release) until 70% of maximum thrust (the highest value of thrust) is achieved on the initial rise portion of the thrust-time curve.

6.3.2 **Peak thrust.** Peak thrust is the maximum value of thrust that occurs after the ignition peak thrust time interval.

6.3.3 **Ignition peak thrust.** Ignition peak thrust is the value of thrust that occurs within the first .030 second after the first indication of thrust.

6.3.4 **Torque.** Torque is measured during the rocket motor burn (time after the ignition peak thrust time interval).

6.4 **Changes from previous issue.** Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

*(See Instructions – Reverse Side)*

| 1. DOCUMENT NUMBER | 2. DOCUMENT TITLE |
|---|---|
| | |

| 3a. NAME OF SUBMITTING ORGANIZATION – | 4. TYPE OF ORGANIZATION (Mark one) |
|---|---|
| | ☐ VENDOR |
| | ☐ USER |
| b. ADDRESS (Street, City, State, ZIP Code) | ☐ MANUFACTURER |
| | ☐ OTHER (Specify): _____ |

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

| 7a. NAME OF SUBMITTER (Last, First, MI) – Optional | b. WORK TELEPHONE NUMBER (Include Area Code) – Optional |
|---|---|
| c. MAILING ADDRESS (Street, City, State, ZIP Code) – Optional | 8. DATE OF SUBMISSION (YYMMDD) |

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